

Curriculum Vitae

Shai Tsesses, Ph.D.

Summary

My research explores a new platform for quantum science and technology, where atoms, free electrons, and photons all interact coherently. I aim to overcome the limitations of current quantum optical technologies and utilize free electron beams as quantum interconnects to scale up leading quantum computing modalities. I am also interested in developing new techniques for quantum computation and simulation with atomic arrays, as well as for electron microscopy.

RESEARCH EXPERIENCE

- 2027-present **Assistant Professor in Electrical and Computer Engineering**, University of Wisconsin – Madison, Madison, WI, USA
- 2023-2027 **Post-Doctoral Fellow**, MIT, Cambridge, MA, USA
 Advisor: Vladan Vuletić
- Made core contributions to experiments improving key functions of the atom array platform, including atom imaging and loading compatible with continuous array operation and fast atom detection via Rydberg atom-polariton interactions
 - Actively participated in designing and building one of the first continuously-operated atom array systems, while overseeing 6 graduate students.
- 2016-2023 **Doctoral candidate**, Technion – Israel Institute of Technology, Israel
 Advisor: Guy Bartal
- Initiated and was involved end-to-end in design, fabrication, theory and experiments in quantum and topological plasmonics, including the first observation of optical skyrmions and entanglement in the total angular momentum of light
 - Initiated and was involved end-to-end in design, fabrication, theory and experiments in quantum free-electron–light interaction, including photo-induced spatial modulation of free-electrons and ultrastrong electron-photon coupling

EDUCATION & TRAINING

- 2023-2027 **Post-Doctoral Fellow**, MIT-Harvard Center for Ultracold Atoms and Research Laboratory of Electronics, MIT, Cambridge, MA, USA
- 2016-2023 **Doctor of Philosophy in Electrical Engineering**, Technion – Israel Institute of Technology, Israel
 Thesis title: “Topology and Angular Momentum Transfer Between Light and Matter in Nanoscale Photonic Systems”
- 2012-2016 **Bachelor of Science** in Electrical Engineering and in Physics (double B.Sc.), Technion – Israel Institute of Technology, Israel (graduated Cum Laude)

INVITED TALKS & SEMINARS

- 1) ECE Quantum Engineering Seminar, University of Wisconsin – Madison (Madison, WI, USA, 2026). Title: “Unlocking New Capabilities for Quantum Computation with Neutral Atom Arrays”
- 2) Physics & Astronomy Colloquium, Rice University (Houston, TX, USA, 2026). Title: “Unlocking New Capabilities for Quantum Computation with Neutral Atom Arrays”
- 3) School of Physical and Mathematical Sciences Seminar, Nanyang Technological University (Singapore, Singapore, 2026). Title: “Topology and new interaction regimes between light and matter in nanoscale photonic systems”
- 4) Electro-Optics and Microelectronics Seminar, Technion – Israel Institute of Technology (Haifa, Israel, 2026). Title: “Unlocking New Capabilities for Quantum Computation with Neutral Atom Arrays”
- 5) Atomic, Molecular and Optical Sciences Seminar, the Weitzmann institute (Rehovot, Israel, 2026). Title: “Unlocking New Capabilities for Quantum Computation with Neutral Atom Arrays”
- 6) Quantum Optics Resnick Seminar, Bar-Ilan University (Ramat Gan, Israel, 2026). Title: “Unlocking New Capabilities for Quantum Computation with Neutral Atom Arrays”
- 7) Condensed Matter Seminar, Tel-Aviv University (Tel-Aviv, Israel, 2026). Title: “Unlocking New Capabilities for Quantum Computation with Neutral Atom Arrays”

- 8) 10th International Topical Meeting on Nanophotonics and Metamaterials – NANOMETA-2026 (Seefeld, Austria, 2026). Title: “Rydberg atom-polariton interaction: exceptional points, nonlinear networks and remote readout”
- 9) Quantum Materials and Sensing Institute Seminar, Northeastern University (Burlington, MA, USA, 2025). Title: “Quantum optics with Rydberg atoms and free electrons”
- 10) Quantum Control of Light and Matter – Gordon Research Seminar (Newport, RI, USA, 2025). Title: “Topological and Quantum Plasmonics with the Spin-Orbit Interaction of Light”
- 11) Institute for Quantum Optics and Quantum Information Seminar @ University of Innsbruck (Innsbruck, Austria, 2024). Title: “Quantum Computing and Sensing with Free Electrons in Cavity QED Systems”
- 12) 9th International Topical Meeting on Nanophotonics and Metamaterials – NANOMETA-2024 (Seefeld, Austria, 2024). Title: “4D Conserved Topological Charge Vectors in Plasmonic Quasicrystals”
- 13) 4th Physics Institute Seminar, University of Stuttgart (Stuttgart, Germany, 2022). Title: “Revealing Topological and Thermodynamic Properties of Quasicrystals Through Plasmonic Quasi-Lattices”
- 14) Pritzker School of Molecular Engineering Special – Quantum Seminar @ University of Chicago (Chicago, IL, USA, 2022). Title: “Engineering Plasmonic Fields for Quantum Free-Electron-Light Interactions and Novel Light Topologies”
- 15) Atomic, Molecular and Optical Sciences Seminar, the Weitzmann institute (Rehovot, Israel, 2022). Title: “Free-Electron Interaction with Engineered Plasmonic Fields”
- 16) Q-SORT – International Conference on Quantum Imaging and Electron Beam Shaping (Online, 2021). Title: “Tunable Photo-Induced Free-Electron Spatial Modulation”
- 17) Applied Math Seminar, Tel Aviv University (Tel Aviv, Israel, 2020). Title: “Topology, Polygons and Electromagnetic Surface Waves”
- 18) Minerva-Gentner Symposium on Quantum Simulations using Photons, Atoms, and Molecules (Tze’elim, Israel, 2020). Title: “Optical Skyrmions - a New Texture of Light”.
- 19) Q-SORT – International Conference on Quantum Imaging and Electron Beam Shaping (Erlangen, Germany, 2019). Title: “Dihedral Plasmonics: From Optical Skyrmions to Novel Spin-Orbit Interaction of Light”

TEACHING & MENTORING EXPERIENCE

Student Supervision and Mentoring

12 B.S. students, 12 Ph.D. students (as a co-researcher) 2018-2027

Head Teaching Assistant – Undergraduate Courses

Nano-photonics and optical metamaterials (Technion, Israel; 10-20 students) 2019-2021

Developed the entire course material together with lecturer, gave recitation lectures, wrote exams and HW

Electro-optic Systems (Technion, Israel; 20-40 students) 2016-2018

Developed new course material together with lecturer, gave recitation lectures, wrote exams and HW

Teaching Assistant – Undergraduate Courses

Semiconductor Lasers (Technion, Israel; 15 students) 2017

HW check, exam question writing

Waves and Distributed Systems (Technion, Israel; 80 students) 2016

HW check, exam question writing

AWARDS & FELLOWSHIPS

Yad Hanadiv Foundation's **Rothschild Post-Doctoral Fellowship** (2023)

CHE-PBC Post-Doctoral Fellowship in Quantum Science and Technology (2022,2023)

Helen Diller Post-Doctoral Fellowship in Quantum Science and Technology (2022)

Viterbi Fellowship for Nurturing Future Faculty Members in ECE (2022)

Best Poster Award, WE-Heraeus Seminar on Quantum Electron Optics (2022)

Yablonovich Research Prize (2020)

Adams Fellowship of the Israel Academy of Sciences and Humanities (2019)

OPTICA **Tingye Li Innovation Prize** (2019)

Russel Berrie Prize – 1st place (2019)

Jacobs Foundation Fellowship for Graduate Students (2018)

KLA-Tencore Prize for Excellent Conference Papers (2018)

Russell Berrie Fellowship for MSc students (2017) and PhD students (2019)

Meyer Foundation Award for Graduate Students (2016)
Meyer Fellowship for Undergraduate Students (2012-2016)

MILITARY SERVICE, PROFESSIONAL OUTREACH & COMMUNITY SERVICE

2026	Program Co-Chair , QuARC 2026 conference
2025-present	Grant Referee , Gordon & Betty Moore Foundation
2017-present	Referee , employed by: Optics Letters, Optics Express, Photonics Research, Optica (OPTICA); Physical Review A, Physical Review B, Physical Review Applied, Physical Review Research, Physical Review Letters, Physical Review X (APS); New Journal of Physics (IOP); Nature Communications, Nature Physics, Nature, Nature Materials, Nature Nanotechnology (NPG); ACS Photonics, Nano Letters (ACS); Science Advances (AAAS); PNAS (NAS); Applied Physics Letters (AIP)
2016-2022	Officer , Technion OPTICA chapter [President ('17-'19); VP ('16,'20-'22)]
2007-2011	Israel Defense Forces (mandatory service + officer's course)

Publications

1. A. Kam, C. Roques-Carmes, **S. Tsesses**, A. Karnieli, "Quantum Skyrmions in Mixed States of Light and their Nested Topology", ACS Photonics **13**, 5425-5433 (2026).
2. A. Kam, K. Sulimany (equal contribution), **S. Tsesses**, U. Pereg, "Reduced State Embedding for Error Correction in Quantum Cryptography", Newton **2**, 100597 (2026)
3. E. Qiu, T. Sumarac (equal contribution), P. Niu (equal contribution), **S. Tsesses**, F. Wassaf, D. C. Spierings, M.-W. Chen, M. T. Uysal, A. Bartlett, A. Menssen, M. Lukin, V. Vuletic, "Loading and Imaging Atom Arrays via Electromagnetically Induced Transparency", PRX Quantum **7**, 020358 (2026)
4. A. Kam, **S. Tsesses**, L. Fridman, Y. Ilin, K. Cohen, A. Sivan, L. Popilevsky, M. Orenstein, M. Segev and G. Bartal, "Quantum skyrmions and high dimensional entanglement mediated by nanophotonics", eLight **6**, 13 (2026)
5. T. Bucher, A. Gorlach, A. Niedermayr, Q. Yan, H. Nahari, K. Wang, R. Ruimy, Y. Adiv, M. Yannai, T. L. Abudi, C. Spaegele, C. Roques-Carmes, E. Janzen, J. H. Edgar, F. H. L. Koppens, G. M. Vanacore, H. H. Sheinfux, **S. Tsesses**, I. Kaminer, "Ultrafast Correlations in Ensembles of Optical Phase Singularities", Nature **651**, 920-926 (2026)
6. S. Tsesses & A. Karnieli, "Electrically tuned light topology", Nat. Phys. **21**, 1869-1870 (2025)
[News & Views]
7. F. J. García de Abajo et al, "Roadmap for Photonics with 2D Materials", ACS Photonics **12**, 3961-4095 (2025)
8. J. Schwab, A. Neuhaus, K. Cohen, P. Dreher, **S. Tsesses**, A. Mantha, F. Mangold, B. Frank, G. Bartal, F.-J. Meyer zu Heringdorf, T. J. Davis, H. Giessen, "Skyrmion Bags of Light in Plasmonic Moiré Superlattices", Nat. Phys. **21**, 988-994 (2025)
9. A. Kam, **S. Tsesses (equal contribution)**, Y. Ilin, K. Cohen, Y. Lumer, L. Fridman, S. Lotan, A. Patsyk, L. Nemirovsky-Levy, M. Orenstein, M. Segev, G. Bartal "Near-field photon Entanglement in Total Angular Momentum", Nature **640**, 634-640 (2025)
10. S. Izhak, A. Karnieli, O. Yesharim, **S. Tsesses**, A. Arie, "Pseudospin Localization of Light in Nonlinear Optics", Phys. Rev. Lett. **134**, 123803 (2025)
11. **S. Tsesses**, P. Dreher, D. Janoschka, K. Cohen, T. C. Meiler, T. Bucher, S. Sapir, B. Frank, T. J. Davis, F. Meyer zu Heringdorf, H. Giessen, G. Bartal, "Four-dimensional conserved topological charge vectors in plasmonic quasicrystals", Science **387**, 644-648 (2025)
12. J. Kheireddine, K. Cohen, K. Frischwasser, B. Gjonaj, **S. Tsesses**, G. Bartal, "Dynamic control and manipulation of near-fields using direct feedback", Light Sci. Appl. **13**, 298 (2024)
13. D. Roitman, A. Karnieli, **S. Tsesses**, Z. Barkay, A. Arie, "Coherent cathodoluminescence at visible wavelengths from sub-keV electrons", Opt. Lett. **49**, 2013-2016 (2024)
14. T. Bucher, H. Nahari, H. H. Sheinfux, R. Ruimy, A. Niedermayr, R. Dahan, Q. Yan, Y. Adiv, M. Yannai, J. Chen, Y. Kurman, S. T. Park, D. J. Masiel, E. Janzen, J. H. Edgar, F. Carbone, G. Bartal, **S. Tsesses**, F. H. L. Koppens, G. M. Vanacore, I. Kaminer, "Coherently amplified ultrafast imaging in a free-electron interferometer", Nat. Photon. **18**, 809-815 (2024)
15. A. Karnieli, **S. Tsesses (equal contribution)**, R. Yu, N. Rivera, A. Arie, I. Kaminer, S. Fan, "Universal and ultrafast quantum computation based on free-electron-polariton blockade", PRX Quantum **5**, 010339 (2024)

16. S. Izhak, A. Karnieli, O. Yesharim, **S. Tsesses**, A. Arie, "All-optical spin-valve effect in nonlinear optics", *Opt. Lett.* 49, 1025-1028 (2024)
17. M. Iluz, K. Cohen, J. Kheireddine, Y. Hazan, A. Rosenthal, **S. Tsesses**, G. Bartal, "Unveiling the evolution of light within photonic integrated circuits", *Optica* 11, 42-47 (2024)
18. T. Bucher, R. Ruimy, **S. Tsesses**, R. Dahan, G. Bartal, G. M. Vanacore, I. Kaminer, "Free-Electron Ramsey-Type Interferometry for Enhanced Amplitude and Phase imaging of Nearfields", *Sci. Adv.* 9, eadi5729 (2023)
19. N. Nishad, A. Keselman, T. Lahaye, A. Browaeys, **S. Tsesses**, "Quantum simulation of generic spin exchange models in Floquet-engineered Rydberg atom arrays", *Phys. Rev. A* 108, 053318 (2023)
20. K. Cohen, **S. Tsesses**, S. Dolev, Y. Blechman, G. Ankonina, G. Bartal, "Topological transitions and surface umklapp scattering in weakly-modulated periodic metasurfaces", *Nano Letters* 23, 10243-10250 (2023)
21. Y. Blechman, **S. Tsesses**, M. Feinstein, G. Bartal, E. Almeida, "Room-temperature strong coupling at the nanoscale achieved by inverse design", *ACS Photonics* 10 (8), 2494-2501 (2023)
22. **S. Tsesses**, R. Dahan (equal contribution), K. Wang, T. Bucher, K. Cohen, O. Reinhardt, G. Bartal, I. Kaminer, "Tunable photon-induced spatial modulation of free electrons", *Nat. Mater.* 22, 345-352 (2023) **[Selected for Cover Image and News & Views]**
23. Y. Adiv, H. Hao (equal contribution), **S. Tsesses (equal contribution)**, R. Dahan, K. Wang, Y. Kurman, H. Chen, X. Lin, G. Bartal, I. Kaminer, "Observation of 2D Cherenkov Radiation", *Phys. Rev. X* 13, 011002 (2023)
24. A. Karnieli, **S. Tsesses (equal contribution)**, R. Yu (equal contribution), N. Rivera, Z. Zhao, A. Arie, S. Fan, I. Kaminer, "Quantum sensing of strongly coupled light-matter systems using free electrons", *Sci. Adv.* 9, eadd234 (2023)
25. Y. Blechman, **S. Tsesses**, G. Bartal, E. Almeida, "Inverse design of broadband, strongly-coupled plexcitonic nonlinear metasurfaces", *New J. Phys.* 24, 095003 (2022)
26. A. Karnieli, D. Roitman (equal contribution), M. Liebtrau (equal contribution), **S. Tsesses (equal contribution)**, N. Van Nielen, I. Kaminer, A. Arie, A. Polman, "Cylindrical Metasurfaces for Generation and Focusing of Free-Electron Radiation", *Nano Lett.* 22, 5641-5650 (2022)
27. K. Frischwasser, K. Cohen, **S. Tsesses**, S. Dolev, G. Rosenblatt, G. Bartal, "Nonlinear forced response of plasmonic nanostructures", *Phys. Rev. Lett.* 128, 103901 (2022)
28. A. Dikopoltsev, Y. Sharabi (equal contribution), M. Lyubarov, Y. Lumer, **S. Tsesses**, E. Lustig, I. Kaminer, M. Segev, "Light Emission by Free Electrons in Photonic Time-Crystals", *Proc. Natl. Acad. Sci. U.S.A* 119, e2119705119 (2022)
29. K. Frischwasser, K. Cohen, Y. Kheralden, S. Dolev, **S. Tsesses**, G. Bartal, "Real-time subwavelength imaging of surface waves with nonlinear near-field optical microscopy", *Nat. Photonics* 15 (6), 442-448 (2021)
30. A. Karnieli, **S. Tsesses**, G. Bartal, A. Arie, "Emulating spin transport with nonlinear optics, from high-order skyrmions to the topological Hall effect", *Nat. Commun.* 12, 1092 (2021)
31. Y. Ilin, **S. Tsesses**, G. Bartal, Y. Sagi, "Sub-wavelength spin excitations in ultracold gases created by stimulated Raman transitions", *New J. Phys.* 22, 093071 (2020)
32. K. Wang, R. Dahan, M. Shentcis, Y. Kauffmann, A. Ben Hayun, O. Reinhardt, **S. Tsesses**, Ido Kaminer, "Coherent interaction between free electrons and a photonic cavity", *Nature* 582 (7810), 50-54 (2020)
33. D. Panna, N. Landau, L. Rybak, L. Gantz, **S. Tsesses**, G. Adler, S. Brodbeck, S. Hoefling, C. Schneider, A. Hayat, "Ultrafast manipulation of a strongly-coupled light-matter system by a giant ac Stark effect", *ACS Photonics* 6 (12), 3076-3081 (2019)
34. **S. Tsesses**, K. Cohen, E. Ostrovsky, B. Gjonaj, G. Bartal, "Spin-orbit interaction of light in plasmonic lattices", *Nano Lett.* 19 (6), 4010-4016 (2019)
35. R. Ravi Kumar, E. Hahamovich, **S. Tsesses**, Y. Hazan, A. Grinberg, A. Rosenthal, "Enhanced sensitivity of silicon-photonics-based ultrasound detection via BCB coating", *IEEE Photonics J.* 11 (3), 1-11 (2019)
36. **S. Tsesses**, E. Ostrovsky, K. Cohen, B. Gjonaj, N. H. Lindner, G. Bartal, "Optical skyrmion lattice in evanescent electromagnetic fields", *Science* 361 (6406), 993-996 (2018)
37. Y. Kurman, N. Rivera, T. Christensen, **S. Tsesses**, M. Orenstein, M. Soljačić, J. D. Joannopoulos, I. Kaminer, "Controlling emitter frequency by increasing photon momentum", *Nat. Photonics* 12 (7), 423-429 (2018)
38. E. Ostrovsky, K. Cohen, **S. Tsesses**, B. Gjonaj, G. Bartal, "Nanoscale control over optical singularities", *Optica* 5 (3), 283-288 (2018)

39. **S. Tsesses**, A. Grinberg, D. Aronovich, E. Hahamovich, A. Rosenthal, "Modeling the sensitivity dependence of silicon-photonics-based ultrasound detectors", Opt. Lett. 42 (24), 5262-5265 (2017)
40. **S. Tsesses**, G. Bartal, I. Kaminer, "Light Generation via Quantum Interaction of Electrons with Periodic Nanostructures", Phys. Rev. A 95 (1), 013832 (2017)

Pre-published works

1. T. Sumarac, E. Qiu (equal contribution), **S. Tsesses (equal contribution)**, M. L. Peters, P. Niu, A. Menssen, V. Walther, U. Delic, W. Xu, S. Choi, M. Lukin, V. Vuletic, "Controlling Rydberg atom-polariton interactions: from exceptional points to fast readout", Under Review in Physical Review X (2026), arXiv:2601.06345
2. A. J. Menssen et al, "Strategic Plan for Neutral Atom Quantum Computation", Under Review in PRX Quantum, arXiv preprint arXiv:2607.21554
3. S. Even-Haim, D. Gelem, A. Karnieli, R. Yu, I. Kaminer, V. Vuletić, **S. Tsesses**, "Passive Nanophotonic Doppler Cooling of Free Electrons", in preparation
4. J. Bender, **S. Tsesses**, F. Wassaf, G. Baranes, P. Niu, E. H. Qiu, V. Vuletic, S. Choi, "Multi-Qubit Stabilizer Gates via Double-Rydberg Excitation", in preparation

Patents

1. A. Rozental, R. R. Kumar, S. Tsesses, A. Grinberg, "Polymer-coated high-index waveguide for acoustic sensing", U.S. Patent No. 11,619,543 (2023)

Conference Proceedings

1. **S. Tsesses**, G. Bartal, I. Kaminer, "Smith-Purcell radiation in periodic nanostructures: quantum effects and applicability", CLEO: QELS_Fundamental Science, FTu4D. 6 (2017).
2. **S. Tsesses**, K. Cohen, E. Ostrovsky, B. Gjonaj, G. Bartal, "Controlling the topology of plasmonic vortex lattices", CLEO: QELS_Fundamental Science, FTh3M. 4 (2018).
3. D. Panna, N. Landau, S. Bouscher, L. Rybak, **S. Tsesses**, G. Adler, S. Brodbeck, C. Schneider, S. Hoefling, A. Hayat, "Giant AC Stark Effect in a Strongly-Coupled Light-Matter System", CLEO: QELS_Fundamental Science, FM2F. 1 (2018).
4. Y. Kurman, N. Rivera, T. Christensen, **S. Tsesses**, M. Orenstein, M. Soljacic, J. Joannopoulos, I. Kaminer, "Inducing Indirect Optical Transitions Using Graphene Plasmons", CLEO: QELS_Fundamental Science, FF2L. 3 (2018).
5. R. Kolkowski, H. Hertzog-Sheinfux, **S. Tsesses**, A. Kodanov, M. Orenstein, G. Bartal, M. Segev, "Evanescent fields at the gain/loss interface", CLEO: QELS_Fundamental Science, FM2Q. 5 (2018).
6. **S. Tsesses**, D. Aronovich, A. Grinberg, E. Hahamovich, A. Rosenthal, "Modeling of ultrasound detection by silicon-photonics-based sensors", Novel Optical Materials and Applications, JTu2A. 43 (2018).
7. **S. Tsesses**, E. Ostrovsky, K. Cohen, B. Gjonaj, N. H. Lindner, G. Bartal, "Optical skyrmions: a new topological state of light", CLEO: QELS_Fundamental Science, FTu3D. 5 (2019). **[awarded the Tingye Li innovation prize]**
8. Resmi R. Kumar, Evgeny Hahamovich, **Shai Tsesses**, Yoav Hazan, Assaf Grinberg, Amir Rosenthal, "Sensitivity enhancement of silicon-photonics-based ultrasound sensor via BCB cladding", CLEO: Science and Innovations, SW3L. 3 (2019).
9. Resmi R. Kumar, Evgeny Hahamovich, **Shai Tsesses**, Yoav Hazan, Assaf Grinberg, Amir Rosenthal, "Hybrid ultrasound-detection platform based on silicon photonics and transparent polymers", Proc. SPIE. 11240, Photons Plus Ultrasound: Imaging and Sensing 2020 (2020).
10. Y. Blechman, **S. Tsesses**, G. Feinberg, A. Hayat, G. Bartal, "Optimizing the strong coupling of excitons in 2D materials and surface plasmon lattice resonances", CLEO: QELS_Fundamental Science, FM1D. 5 (2020).
11. D. Shterman, O. Eyal, **S. Tsesses**, G. Bartal, "Tessellation structured illumination microscopy", CLEO: Applications and Technology, AM2I. 3 (2020).
12. K. Wang, R. Dahan, M. Shentcis, A. Ben-Hayun, O. Reinhardt, **S. Tsesses**, I. Kaminer, "Free electron cavity quantum electrodynamics in an ultrafast electron microscope", CLEO: QELS_Fundamental Science, FTu3D. 6 (2020).

13. O. Reinhardt, K. Wang, **S. Tsesses**, S. Nehemia, R. Dahan, M. Shentcis, I. Kaminer, "Quantum walk with coherent uncertainty in electron-laser interaction", CLEO: QELS_Fundamental Science, FTu3B. 3 (2020).
14. **S. Tsesses**, R. Dahan, K. Wang, G. Bartal, I. Kaminer, "Spatial modulation of free electron wavepackets by shaping ultrafast plasmonic excitations", CLEO: QELS_Fundamental Science, FTu3B. 4 (2020).
15. **S. Tsesses**, K. Cohen, E. Ostrovsky, B. Gjonaj, T. Bucher, S. Sapir, G. Bartal, "Spin-orbit interaction of light in plasmonic lattices: modified and broken angular momentum conservation", CLEO: QELS_Fundamental Science, JTu4A. 3 (2020).
16. D. Shterman, G. Feinberg, **S. Tsesses**, Y. Blau, G. Bartal, "Deep Learning structured illumination microscopy", CLEO: Applications and TEchnology, AW3T. 4 (2020).
17. A. Karnieli, **S. Tsesses**, G. Bartal, A. Arie, "Optical skyrmions and topological hall effect in artificial gauge fields", CLEO: QELS_Fundamental Science, FW4A. 6 (2020).
18. K. Cohen, **S. Tsesses**, Y. Blechman, A. David, G. Bartal, "Aliasurface - aliasing based metasurface", CLEO: QELS_Fundamental Science, FTh4B. 1 (2020).
19. A. Dikopoltsev, Y. Sharabi, **S. Tsesses**, I. Kaminer, M. Segev, "Free-electrons radiation in a photonic time crystal", CLEO: QELS_Fundamental Science, FF1Q. 4 (2020).
20. **S. Tsesses**, R. Dahan, K. Wang, O. Reinhardt, G. Bartal, I. Kaminer, "Active spatial modulation of free electrons by controlling the shape and intensity of plasmonic fields", CLEO: QELS_Fundamental Science, FM1L. 2 (2021).
21. Y. Adiv, H. Hu, **S. Tsesses**, R. Dahan, K. Wang, Y. Kurman, H. Chen, X. Lin, G. Bartal, I. Kaminer, "Observation of 2D Cherenkov radiation and its quantized photonic nature using free-electrons", CLEO: QELS_Fundamental Science, FM1L. 6 (2021).
22. K. Frischwasser, K. Cohen, J. Kher-Alden, S. Dolev, **S. Tsesses**, G. Bartal, "Real-time imaging of surface waves with nonlinear near-field optical microscopy", CLEO: QELS_Fundamental Science, FW2L. 5 (2021).
23. Y. Ilin, **S. Tsesses**, G. Bartal, Y. Sagi, "Sub-wavelength spin excitations driven by stimulated Raman transitions in ultracold gases", CLEO: QELS_Fundamental Science, FW3P. 6 (2021).
24. A. Karnieli, **S. Tsesses**, I. Kaminer, G. Bartal, A. Arie, "Nonlinear optical spintronics: topological Hall effect and Anderson localization", CLEO: QELS_Fundamental Science, FTh1J. 2 (2021).
25. Y. Blechman, **S. Tsesses**, E. Almeida, G. Bartal, "Nonlinear amplitude and spectral control in strongly coupled plasmonic-excitonic nanostructures", CLEO: QELS_Fundamental Science, FTh2K. 3 (2021).
26. A. Karnieli, D. Roitman, M. Liebrau, **S. Tsesses**, N. Van Nielsen, A. Polman, I. Kaminer, A. Arie, "Smith-Purcell metasurface lens", CLEO: QELS_Fundamental Science, JTh3A. 32 (2021).
27. A. Dikopoltsev, Y. Sharabi, Y. Lumer, M. Lyubarov, **S. Tsesses**, E. Lustig, I. Kaminer, M. Segev, "Light emission by free electrons in photonic time-crystals", CLEO: QELS_Fundamental Science, FF2H. 5 (2021).
28. Y. Adiv, H. Hu, **S. Tsesses**, R. Dahan, K. Wang, Y. Kurman, H. Chen, X. Lin, G. Bartal, I. Kaminer, "Achieving Single-Electron-Single-Photon Interaction in a Transmission Electron Microscope Using 2D Cherenkov Radiation", Proceedings of Electron Beam Spectroscopy for Nanooptics 2021 (EBSN2021) (2021).
29. A. Karnieli, D. Roitman, M. Liebrau, **S. Tsesses**, N. Van Nielsen, A. Polman, I. Kaminer, A. Arie, "Smith-Purcell metasurface lens", Proceedings of Electron Beam Spectroscopy for Nanooptics 2021 (EBSN2021) (2021).
30. A. Karnieli, D. Roitman, M. Liebrau, **S. Tsesses**, N. van Nielsen, I. Kaminer, A. Arie, A. Polman, "Focusing free-electron radiation with source-embedded metalenses", CLEO: QELS_Fundamental Science, FM4F. 2 (2022).
31. A. Karnieli, **S. Tsesses**, R. Yu, N. Rivera, Z. Zhao, A. Arie, S. Fan, I. Kaminer, "Probing strongly coupled light-matter interactions using quantum free electrons", CLEO: QELS_Fundamental Science, FTh5L. 4 (2022).
32. **S. Tsesses**, A. Keselman, A. Browaeys, T. Lahaye, "Generating an arbitrary spin exchange between Rydberg atoms through Floquet engineering", Quantum 2.0, QW3A. 3 (2022).
33. N. Nishad, A. Keselman, **S. Tsesses**, "Simulating generic spin exchange models in Rydberg atom arrays via Floquet engineering", Bulletin of the American Physical Society (2023).
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